HANNA PARK

Computer Engineering

Embedded Firmware · RTOS · RTL/ASIC · Embedded Linux

University of Waterloo · BASc Computer Engineering, Expected 2028 · h93park@uwaterloo.ca

PROFILE

I work close to hardware: embedded C/C++, real-time systems, SystemVerilog RTL, Embedded Linux, motor and sensor interfaces, hardware debugging, and validation. My strongest work spans the boundary between software, digital hardware, and physical systems.

TECHNICAL FOCUS

Embedded firmware	ATSAMD51, STM32, UART/SPI/I²C, drivers, sensors, motor control
Real-time systems	Cortex-M4, SVC/PendSV, task state, heap management, EDF scheduling
RTL / digital hardware	SystemVerilog, fixed-point ALU, FPGA/ASIC design, pipelining
Embedded Linux	Yocto, U-Boot, device trees, systemd, networking, SSH, provisioning

SELECTED WORK

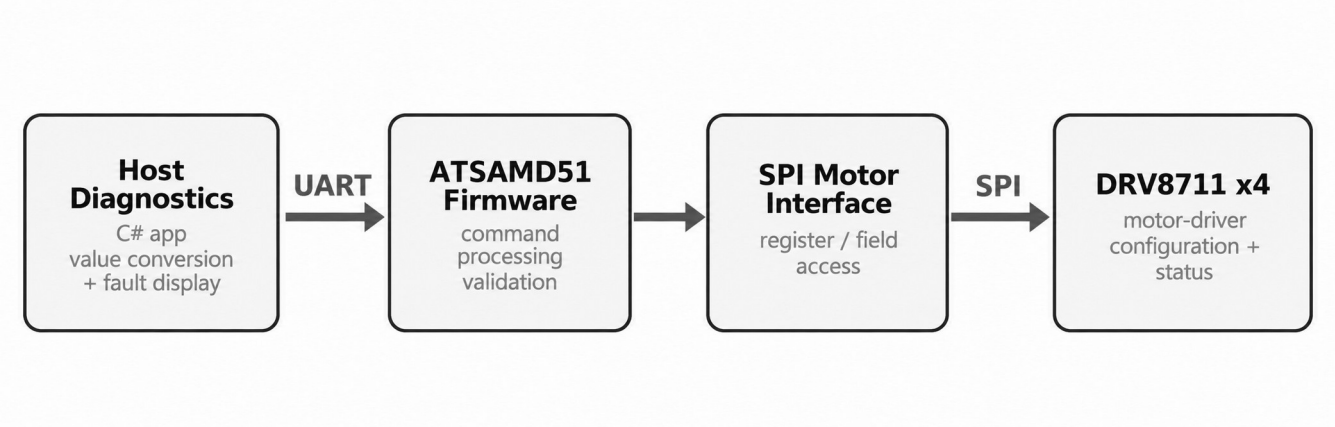
01	**Qidni Labs - Embedded Control & Diagnostics**	firmware, motor/valve drivers, UART/SPI, hardware debugging
02	**Qidni Labs - Embedded Linux Platform**	Yocto, U-Boot, device trees, networking, provisioning
03	**Qidni Labs - Sensor Validation & Physical Integration**	controlled testing, root cause, rapid prototyping
04	**Real-Time & Digital Hardware**	ECE 350 RTOS, UWASIC TinyTracer, ECE 327 FPGA
05	**Production Software & C/C++ Systems**	DF/Net C++/Qt, multithreaded systems programming
06	**Embedded Control & Experimental Engineering**	STM32 control, FATFS, controls and instrumentation
07	**Education & Additional Experience**	WAT.ai, private tutoring, technical toolkit

Qidni Labs - Embedded Control & Diagnostics

Software Engineer Intern · May-Aug 2026 · Embedded firmware / hardware integration

CONTEXT

Qidni Labs develops medical-device systems that combine fluid handling, embedded electronics, sensors, and control software. My firmware work connected an ATSAMD51 device, a C# host diagnostics application, and SPI motor/valve drivers.



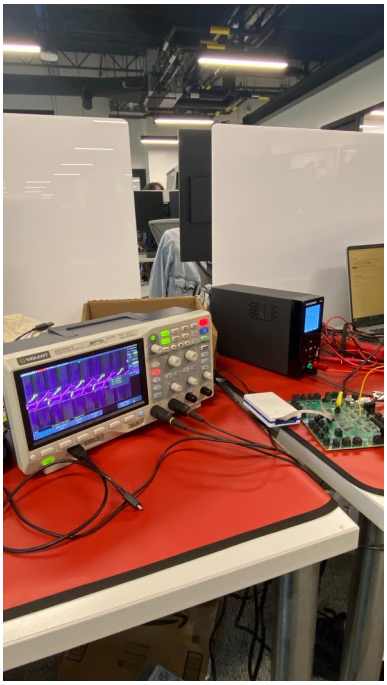
Generalized host → UART → ATSAMD51 → SPI → motor-driver architecture.

MY CONTRIBUTION

- Extended ATSAMD51 firmware and host diagnostics to expose DRV8711 registers and tuning fields across four pump channels, with validated read/write handling and decoded status faults.
- Redesigned the UART command/response path so firmware returned compact raw values while the C# host handled conversion and display formatting.
- Refactored dual DRV81004-Q1 valve control around per-driver PCB mappings and added structured SPI diagnostic decoding for driver fault states.

ENGINEERING FINDING

During a related stall-detection investigation, datasheet review and oscilloscope work showed that the existing PCB did not route the DRV8711 BEMF signal to an ADC-capable MCU input. External BEMF-based detection therefore required a hardware change rather than additional firmware alone.



Motor-control / diagnostics bench work.



Oscilloscope measurement during stall investigation.

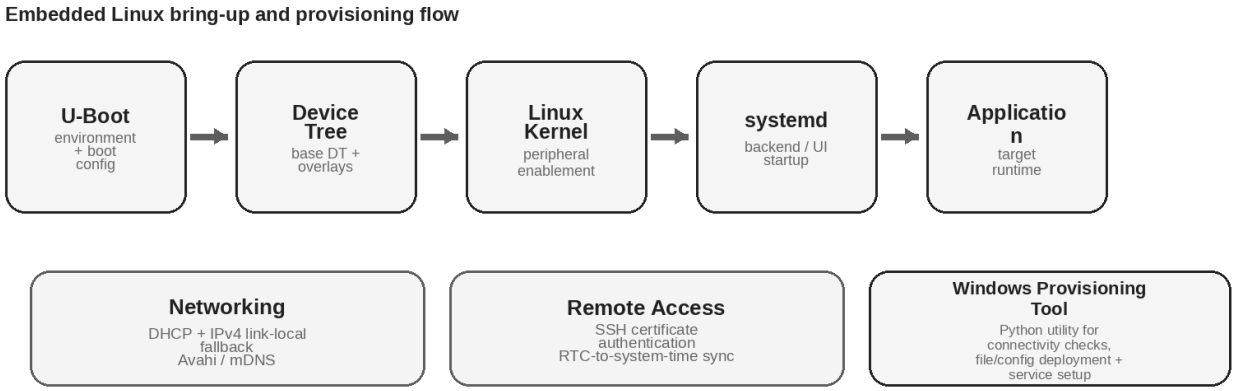
HANNA PARK | COMPUTER ENGINEERING | h93park@uwaterloo.ca

Qidni Labs - Embedded Linux Platform

Yocto · i.MX8M Plus · U-Boot · device trees · systemd · networking · provisioning

CONTEXT

A newly programmed i.MX8M Plus-based SoM needed a repeatable path from bootloader configuration to working peripherals, network access, secure remote access, and automatic application startup.



Generalized boot, deployment, networking, and provisioning flow.

BOARD BRING-UP

- Configured U-Boot environment settings, selected the base device tree, compiled/deployed custom overlays, and installed target binaries and configuration.
- Created and enabled systemd services for backend communication and UI processes, validating boot from U-Boot through Linux to automatic application startup.
- Worked with UART, SPI, I²C, HDMI, and other board-level interfaces during system bring-up.

NETWORKING & REMOTE ACCESS

- Stabilized direct host-to-device Ethernet using DHCP with IPv4 link-local fallback and mDNS.
- Configured certificate-authority SSH authentication and resolved certificate failures caused by incorrect device time by synchronizing RTC time at boot.

PROVISIONING AUTOMATION

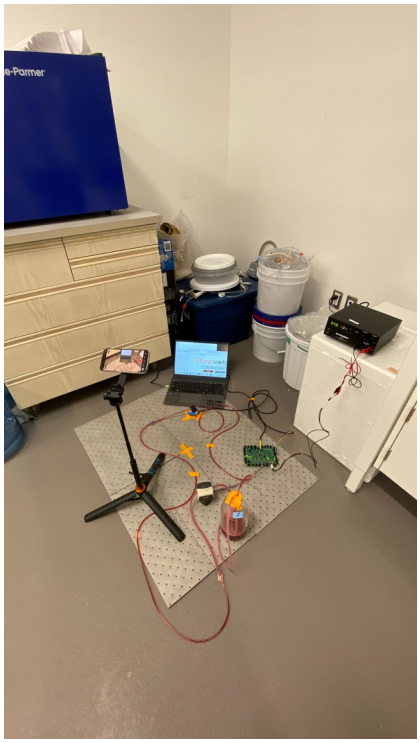
- Built a Python Windows utility to automate connectivity checks, hostname/configuration setup, file deployment, device-tree overlays, scripts, services, and password/certificate SSH workflows for newly programmed boards.

Qidni Labs - Sensor Validation & Physical Integration

Firmware integration · controlled testing · root-cause reasoning · rapid prototyping

AIR-BUBBLE DETECTOR INTEGRATION & VALIDATION

- Ported existing sensor logic to ATSAMD51 platform GPIO APIs and integrated the module into the firmware build.
- Developed repeatable validation tests using controlled tubing, injected bubbles, multiple flow conditions, video-based response-time analysis, and structured test workbooks.
- Used raw sensor observations, video timing, and controlled conditions to distinguish sensor response from physical bubble travel and identify cases requiring re-verification.



Controlled air-bubble detector test setup.



Video/timing setup used during response analysis.

PRESSURE-SENSOR FIXTURE

Designed a simple 3D-printed alignment jig using basic CAD to hold a custom membrane consistently against a pressure sensor during integration work.



3D-printed membrane alignment fixture.

ADDITIONAL VALIDATION / INTEGRATION

- Developed formal verification procedures for net fluid removal and related fault behavior with defined setup, operating conditions, expected measurements, and acceptance checks.
- Investigated fluid-channel leakage by comparing sealing, fit, flow restriction, mechanical strength, and manufacturing tradeoffs across multiple physical approaches.

HANNA PARK | COMPUTER ENGINEERING | h93park@uwaterloo.ca

Real-Time & Digital Hardware

ECE 350 RTOS kernel · UWASIC TinyTracer · ECE 327 FPGA

ECE 350 - REAL-TIME EXECUTIVE / RTOS KERNEL

ARM Cortex-M4 · C / ARM assembly · STM32F4 · four-person course project

The cumulative Real-Time Executive progressed from task creation and cooperative multitasking to kernel memory management and deadline-based preemptive scheduling. Because this was a team project, the portfolio describes the team-developed kernel without assigning unverified ownership of every subsystem.

Application tasks	Task entry points / stacks
SVC-protected kernel APIs	TCBs / task state
First-Fit heap allocator	Metadata / ownership / coalescing
EDF scheduler	Ready / sleeping tasks
PendSV + SysTick	Context switching / timing

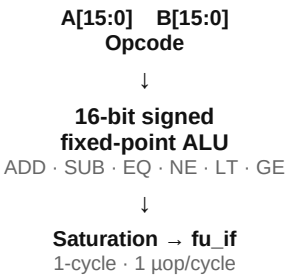
- Task creation/termination, TCB management, kernel startup/yielding, and task information APIs.
- Kernel operations behind SVC exceptions; context switching through PendSV.
- First-Fit heap allocator with block metadata, freelist management, task ownership, four-byte alignment, and immediate coalescing.
- SysTick-driven preemption and Earliest Deadline First scheduling for ready/sleeping periodic tasks.

UWASIC - TINYTRACER RAY-TRACING ASIC

ALU / RTL Engineer · SystemVerilog · digital logic · computer architecture

- Designed the microarchitecture and SystemVerilog RTL for a 16-bit signed fixed-point ALU supporting ADD, SUB, EQ, NE, LT, and GE.
- Implemented 1-cycle execution with 1 µop/cycle throughput and saturating arithmetic at signed 16-bit limits (0x7FFF / 0x8000).
- Integrated the ALU with the accelerator functional-unit handshake interface (fu_if) and verified arithmetic, comparisons, overflow, and boundary cases through RTL simulation.

ECE 327: paired SystemVerilog/FPGA labs included a fixed-point tanh accelerator using Q2.12 arithmetic, ready/valid backpressure and pipelining, plus matrix-vector accelerator work. No specific achieved clock rate is claimed without a timing report.



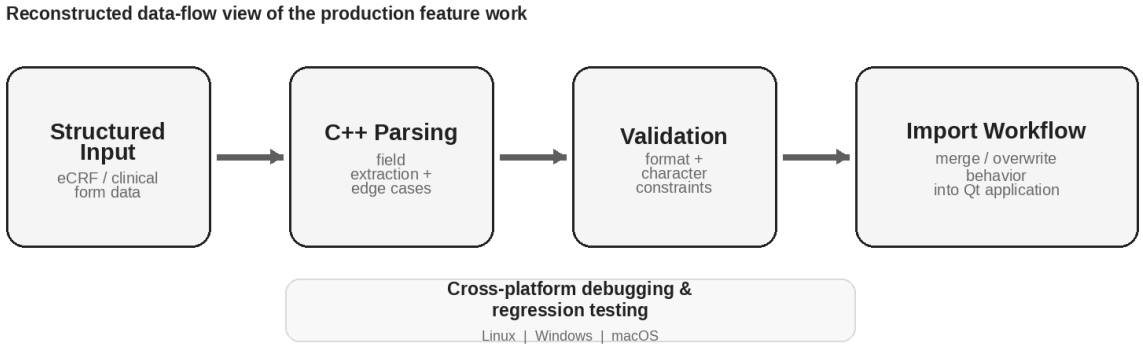
Production Software & C/C++ Systems

DF/Net Research · multithreaded systems programming

DF/NET RESEARCH - C/C++ SOFTWARE DEVELOPER · FEB-AUG 2025

DF/Net develops regulated eClinical software for clinical-trial data collection and management.

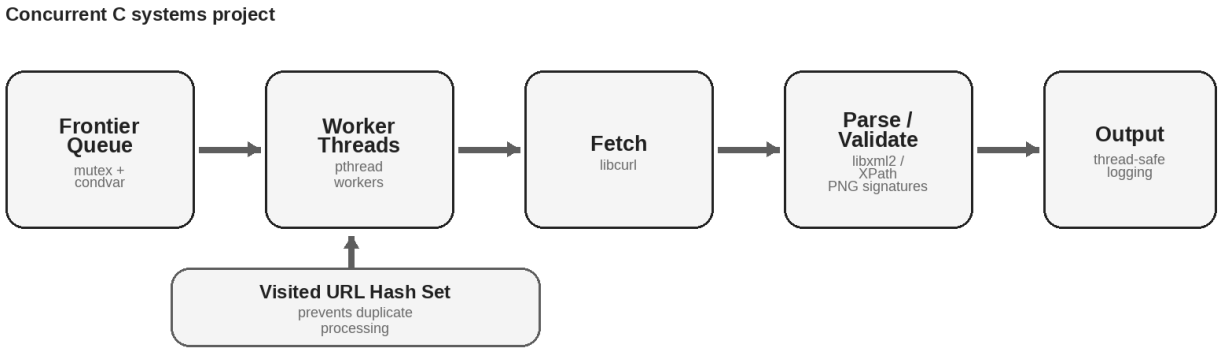
- Developed production C++/Qt functionality, including reusable eCRF/data import with merge/overwrite behavior.
- Implemented structured parsing and field-level validation for high-integrity clinical inputs, including format/character constraints and edge-case handling.
- Diagnosed runtime and parsing issues across Linux, Windows, and macOS and wrote regression/validation tests to preserve predictable behavior.



Generalized structured-data workflow from input through parsing/validation to the Qt application.

MULTITHREADED WEB CRAWLER & BINARY DATA VALIDATOR - C

- Built a thread-safe frontier using pthreads, mutexes, and condition variables plus a visited-URL hash set to avoid duplicate work.
- Used libcurl to fetch pages, libxml2/XPath to extract links, PNG signature validation for binary results, and synchronized logging/output across worker threads.

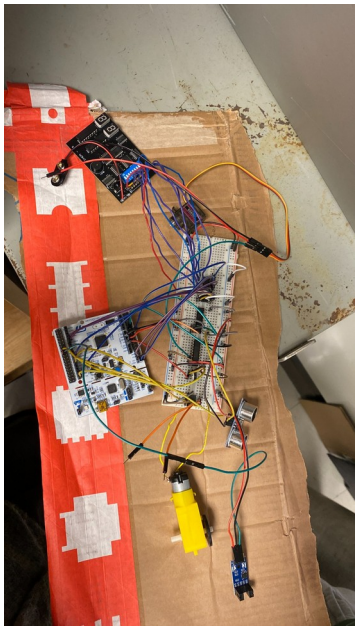


Concurrent work queue → pthread workers → HTTP/XML parsing → validation → synchronized output.

Embedded Control & Experimental Engineering

Complete embedded systems · real-time data paths · controls / instrumentation

WATER RESERVOIR EMBEDDED CONTROL SYSTEM - C / STM32



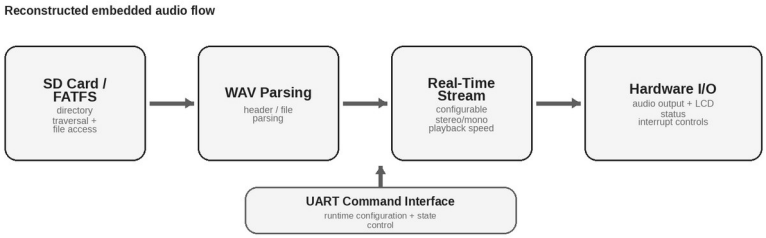
STM32 reservoir prototype during bring-up.

- Built a multi-mode controller integrating PWM motor control, servo actuation, ultrasonic sensing, interrupt-driven RPM measurement, UART configuration, timers, input capture, and empty-reservoir safety shutdown logic.
- Performed system bring-up across firmware, timing, and electrical interfaces and produced a prototype schematic and PCB design.



AUDIO PLAYBACK & FILESYSTEM SYSTEM - C / FATFS

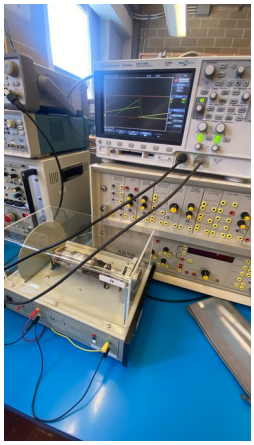
Implemented SD-card/FAT directory traversal, WAV/file parsing, a UART command interface, real-time audio streaming, stereo/mono behavior, variable playback speeds, interrupt-driven controls, debouncing, and display/status feedback.



Generalized FATFS / parsing / audio-streaming flow.

CONTROLS & PHYSICAL SYSTEMS - ECE 380

Hands-on controls labs covered first/second-order system identification, DC motor speed/position control, proportional/PID behavior, root-locus reasoning, and lead/lag compensator design using laboratory instrumentation and oscilloscope measurements.



ECE 380 control-system laboratory work.

HANNA PARK | COMPUTER ENGINEERING | h93park@uwaterloo.ca

Education & Additional Experience

Research, communication, technical breadth, and independent responsibility

---

WAT.AI - MACHINE LEARNING RESEARCHER

May 2025-Dec 2025 · Waterloo, ON

Evaluated vision-language model choices and multimodal fusion strategies for a video-perception pipeline involving frame-level person detection, bounding boxes, and emotion-attribute extraction.

PRIVATE MATHEMATICS & PHYSICS TUTOR - SELF-EMPLOYED

Summer 2023-Present

- Delivered 800+ hours of individualized mathematics and physics instruction to 13 high school students, adapting instruction to different academic goals and learning needs.
- Designed lesson plans, practice assessments, and progressively challenging problem sets; analyzed student performance to identify misconceptions and refine teaching strategies.
- Independently managed scheduling, student assessments, parent communication, and progress reporting.

ADDITIONAL TECHNICAL BACKGROUND

ECE 318	Sampling, aliasing, quantization, PCM encoding/decoding, low-pass reconstruction, and oscilloscope-based measurement.
ECE 222	RISC-V assembly, memory-mapped GPIO, lookup logic, subroutines/stack behavior, and C ↔ assembly interfacing.

EDUCATION

University of Waterloo - BASc Computer Engineering | Expected 2028

- Current coursework: Computer Architecture, Compilers, Computer Networks.
- Selected completed coursework: Real-Time Operating Systems, Digital Hardware Systems, Embedded Microprocessor Systems, Systems Programming & Concurrency, Algorithms & Data Structures, Analog Control Systems, Communication Systems, Instrument & Prototyping Lab.

TECHNICAL TOOLKIT

Languages	C, C++, Python, C#, SystemVerilog, ARM/RISC-V Assembly
Embedded	ATSAMD51, STM32, PWM, timers, input capture, interrupts, UART, SPI, I²C, FATFS, motor drivers
Linux / systems	Linux, Yocto, U-Boot, device trees/overlays, systemd, NetworkManager, mDNS, SSH, pthreads
Hardware / debug	FPGA/RTL, oscilloscope, sensor/actuator bring-up, schematic/PCB work, basic CAD and 3D printing